



SSW3001-2

SOFTWARE ENGINEERING

SEMESTER 1 2025/2026

SOFTWARE DESIGN DOCUMENT

SamaRasa Project

AI-Powered Companion and Health Management System

LECTURER NAME :Tan Ming Ming

Prepared By:

NAME	Matric No.
YUE CHENGHAO(Leader	227154
HUA JIE	226758
WANG KAILUN	227046
ZHANG YAOYUAN	226557
FAN HAOQI	227259
ZHAO YUNUO	227225

Preface

This document contains the system design information for the SamaRasa project. The document is prepared according to the "IEEE Standard for Information Technology – Systems Design – Software Design Descriptions – IEEE Std 1016 – 1998".

This Software Design Documentation provides a complete description of all the system design and views of the SamaRasa Project.

The first section of this document includes purpose, scope, overview, reference material, definitions, and abbreviations of the project.

The second chapter of this document includes an overview of the functionality of the application. It describes the informal design generally.

The third chapter of this document will give the user a detailed description of each function of the system.

The fourth chapter of this document contains data design and data description of the project.

The fifth chapter of this document contains a general overview of what the user interface will look like.

The sixth and last chapter of this document includes the requirements matrix of the project.

Change of History

Version Number	DATE	Number of figure	A,M,D*	Title of Brief Description
3.0	16.01.2026		M	
2.0	14.01.2026		A,D	
1.0	10.01.2026			Original

A.* Added

M.* Modified

D.* Deleted

Table of Contents

1. [Introduction](#)

- 1.1 Scope
- 1.2 Purpose
- 1.3 Overview
- 1.4 Reference Material
- 1.5 Definitions and Abbreviations

2. [Conceptual Model for Software Design Descriptions](#)

- 2.1 Software Design in Context
- 2.2 Software Design Descriptions within the Life Cycle

3. [Design Description Information Content](#)

- 3.1 SDD Identification
- 3.2 Design Stakeholders and Their Concerns
- 3.3 Design Views
- 3.4 Design Viewpoints
- 3.5 Design Rationale
- 3.6 Design Languages

4. [Design Viewpoints](#)

- 4.1 Logical Viewpoint
- 4.2 Information Viewpoint
- 4.3 Interface Viewpoint
- 4.4 Interaction Viewpoint
- 4.5 Context Viewpoints
- 4.6 Composition Viewpoints

5. [Requirements Matrix](#)

Table of Figures

- Figure 1 – Class Diagram of SamaRasa
 - Figure 2 – ER Diagram
 - Figure 3 – System Architecture Diagram
 - Figure 4 – Deployment Diagram
 - Figure 5 – Elderly View Home Screen (English)
 - Figure 6 – Elderly View Home Screen (Malay)
 - Figure 7 – Chat with Me Screen
 - Figure 8 – My Medications Screen
 - Figure 9 – Medication Reminder Screen
 - Figure 10 – Health Diary Screen
 - Figure 11 – Emergency Help Screen
 - Figure 12 – Caregiver Login Screen
 - Figure 13 – Caregiver Dashboard - Medication Schedule
 - Figure 14 – Caregiver Dashboard - Health Analytics
 - Figure 15 – Caregiver Dashboard - Alerts & Logs
 - Figure 16 – Sequence Diagram of AI Chat Component
 - Figure 17 – Sequence Diagram of Medication Reminder Component
 - Figure 18 – Sequence Diagram of Caregiver Add Medication Component
 - Figure 19 – Sequence Diagram of SOS Component
 - Figure 20 – Sequence Diagram of Caregiver Dashboard Component
 - Figure 21 – Sequence Diagram of Mood Recording Component
 - Figure 22 – Sequence Diagram of Data Synchronization
 - Figure 23 – Use Case Diagram of Elderly User
 - Figure 24 – Use Case Diagram of Caregiver
 - Figure 25 – Component Diagram
-

1. Introduction

1.1 Scope

This document provides comprehensive system design information for programmers to develop the SamaRasa application. The document includes all necessary details for code development, including class structures, database schemas, user interfaces, and component interactions.

1.2 Purpose

This document describes the conceptual design of the SamaRasa Project according to the document guidelines presented in the IEEE 1016-1998 Recommended Practice for Software Design Descriptions (SDD).

The SDD shows how the software system will be structured to satisfy the requirements identified in the Software Requirements Specification (SRS). It is a translation of requirements into a description of the software structure, software components, interfaces, and data necessary for the implementation phase. In essence, the SDD becomes a detailed blueprint for the implementation activity. In a complete SDD, each requirement must be traceable to one or more design entities.

1.3 Overview

The purpose of this document is to help the reader visualize the solution to the project presented. This document verifies how the design meets the requirements stipulated in the SRS document through design viewpoints. The design viewpoints will cover all design elements presented before.

By using information from IEEE 1016-1998, this document will provide a direct approach to the development of this project hence reducing feature creep and pointedly determine the quality of the design.

SamaRasa is designed to address Malaysia's aging population challenges by providing:

- Voice-first bilingual interaction (Bahasa Malaysia/English)
- AI-powered emotional companionship
- Smart medication management
- Emergency SOS alerts
- Caregiver monitoring dashboard

1.4 Reference Material

1. IEEE, *IEEE Std 1016-1998 Recommended Practice for Software Design Descriptions*, 1998-09-23, The Institute of Electrical and Electronics Engineers, Inc.
2. SSW3001 Group Project Guidelines, Semester 1 2025/2026, Universiti Putra Malaysia
3. Ministry of Health Malaysia, *National Policy for Older Persons (Revised 2024)*
4. SamaRasa Software Requirements Specification (SRS) Document, Version 1.0

1.5 Definitions and Abbreviations

Term	Definition
Database	Collection of all the information monitored by this system
Elderly User	Primary user of the SamaRasa application, aged 65+
Caregiver	Family member or nurse responsible for monitoring the elderly user
AI	Artificial Intelligence
STT	Speech-to-Text - Technology that converts spoken words to text
TTS	Text-to-Speech - Technology that converts text to spoken words
SOS	Emergency alert system
IEEE	The Institute of Electrical and Electronics Engineers
SRS	Software Requirements Specification
SDD	Software Design Description
JSON	JavaScript Object Notation - A lightweight data-interchange format
HTTPS	Hypertext Transfer Protocol Secure
API	Application Programming Interface
REST/RESTful	Representational State Transfer - An architectural style for APIs
SQLite	A lightweight relational database management system
Firebase	Google's mobile and web application development platform
WCAG	Web Content Accessibility Guidelines
PDPA	Personal Data Protection Act (Malaysia)
UUID	Universally Unique Identifier

2. Conceptual Model for Software Design Descriptions

Information about concepts and context of SDD and stakeholders will be given in this part.

2.1 Software Design in Context

2.1.1 Technologies Used

The system is developed using the following technologies:

Layer	Technology
Frontend (Elderly App)	HTML5, CSS3, JavaScript, React/Vue.js
Frontend (Caregiver Dashboard)	React.js, Responsive Web Design
Backend Server	Node.js / Python (Flask/FastAPI)
Database	SQLite (local) / Firebase Firestore (cloud sync)
Voice Processing	Web Speech API (Browser native STT/TTS)
Communication	HTTPS, JSON, RESTful APIs
Prototyping	Figma

2.1.2 Application Overview

The main goal of the SamaRasa project is to empower elderly Malaysians to live independently while maintaining connection with their caregivers. The final product will be a tablet-optimized web application for elderly users and a responsive web dashboard for caregivers.

Key Features:

1. **AI Chat Companion:** A conversational agent that greets users, provides emotional support, and detects mood through sentiment analysis
2. **Smart Medication Reminder:** Voice-based alerts with visual confirmation for medication adherence
3. **Health Diary:** Tracks daily mood and well-being
4. **Emergency SOS Alert:** One-touch emergency notification system
5. **Caregiver Dashboard:** Remote monitoring and medication schedule management

The application supports bilingual interaction in English and Bahasa Malaysia, with a voice-first interface designed for users with limited digital literacy.

2.2 Software Design Descriptions within the Life Cycle

2.2.1 Influences on SDD Preparation

The Software Requirements Specification (SRS) is the main source for SDD preparation. The functional and non-functional requirements are the main factors determining the design of the project. Key influences include:

- Accessibility requirements for elderly users (WCAG 2.1 compliance)
- Bilingual support requirements
- Real-time synchronization between devices
- Security and privacy requirements (PDPA compliance)

2.2.2 Influences on Software Life Cycle Products

During the implementation and preparation of the SDD, some requirements may change due to unexpected constraints. The testing phase of the project can be prepared using the information in this System Design Document.

2.2.3 Design Verification and Design Role in Validation

Verification determines whether a software work product fulfills specified requirements. Validation determines that the requirements for a specific intended use of a software work product are fulfilled. Therefore, verification and validation results are controlled by the requirements defined in the SRS.

3. Design Description Information Content

3.1 SDD Identification

SamaRasa is designed for Malaysian elderly users aged 65 and above who may have:

- Limited digital literacy
- Visual impairments (presbyopia)
- Motor limitations (hand tremors)
- Cognitive decline (short-term memory issues)

The system will be deployed as a web-based application optimized for tablet devices, with a separate caregiver portal accessible via any modern web browser.

3.2 Design Stakeholders and Their Concerns

Stakeholder	Primary Concerns
Elderly Users	Simplicity, accessibility, reliability, voice interaction

Stakeholder	Primary Concerns
Caregivers	Real-time monitoring, easy configuration, alert notifications
Development Team	Maintainability, scalability, code quality
Healthcare Providers	Data accuracy, compliance with health guidelines

3.3 Design Views

Unified Modeling Language (UML) 2.0 is used for graphical representations of viewpoints in the SamaRasa Project. The following diagram types are provided:

- Class Diagrams
- Entity-Relationship Diagrams
- Sequence Diagrams
- Use Case Diagrams
- Component Diagrams
- Deployment Diagrams

3.4 Design Viewpoints

The following design viewpoints are covered in this document:

- **Logical Viewpoint:** Class structure and relationships
- **Information Viewpoint:** Data entities and database schema
- **Interface Viewpoint:** User interface designs
- **Interaction Viewpoint:** Sequence diagrams showing component interactions
- **Context Viewpoint:** Use case diagrams
- **Composition Viewpoint:** Component architecture

3.5 Design Rationale

The design focuses on:

1. **Accessibility:** Large fonts (min 24px), high contrast (7:1 ratio), large touch targets
2. **Simplicity:** Maximum 2-layer navigation, grid-based layouts
3. **Reliability:** Offline capability with automatic sync, 99.9% medication alarm success rate
4. **Security:** HTTPS encryption, PIN-protected settings, PDPA compliance

3.6 Design Languages

- UML 2.0 for all structural and behavioral diagrams
- Figma for user interface prototypes
- JSON for API data formats

4. Design Viewpoints

4.1 Logical Viewpoint

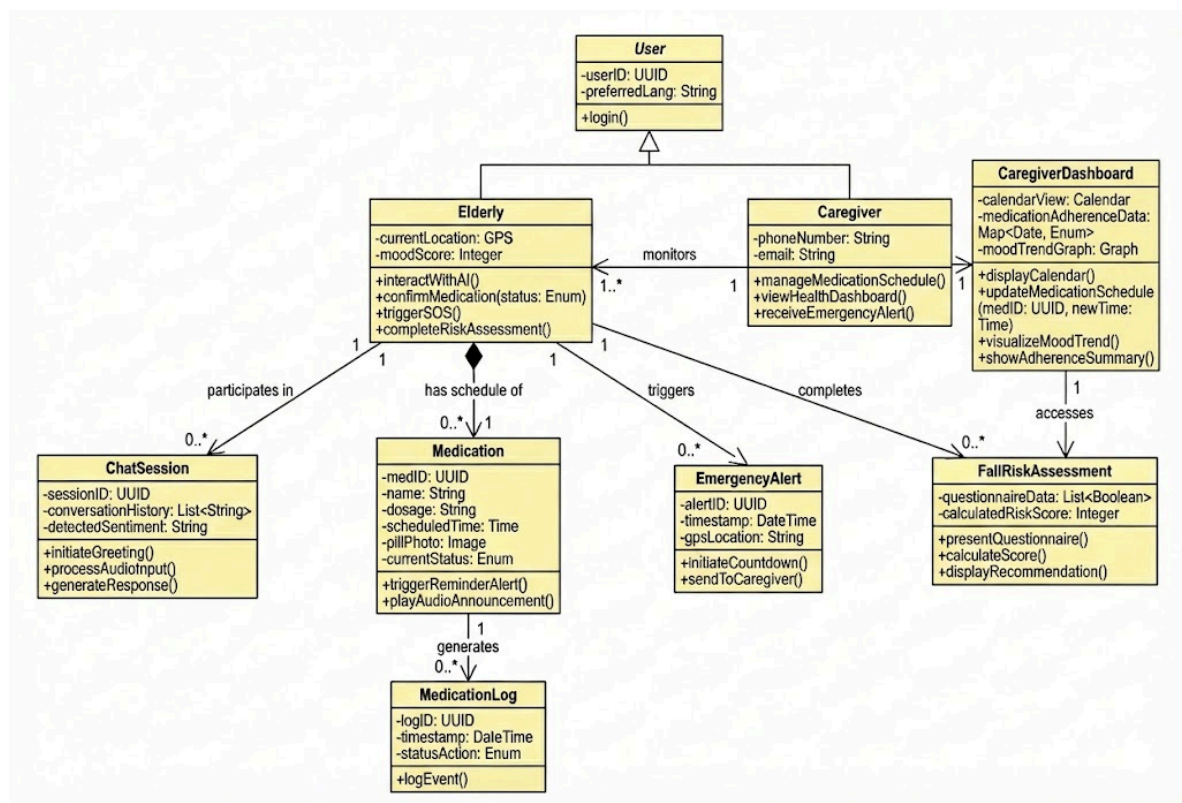
This viewpoint aims to show the key abstractions such as classes and interactions among them.

4.1.1 Design Concerns

The Logical viewpoint is used to address the development and reuse of adequate abstractions and their implementations.

4.1.2 Design Elements - Class Diagram

Figure 1 – Class Diagram of SamaRasa



The class diagram shows the following main classes and their relationships:

User (Abstract Class)

Attribute	Type	Description
userID	UUID	Unique identifier for the user
preferredLang	String	Language preference ('en' or 'ms')

Method	Description
+login()	Authenticates the user

Elderly (extends User)

Attribute	Type	Description
currentLocation	GPS	Current GPS coordinates
moodScore	Integer	Current mood score (0-10)

Method	Description
+interactWithAI()	Initiates conversation with AI companion
+confirmMedication(status: Enum)	Confirms medication taken/skipped
+triggerSOS()	Triggers emergency alert
+completeRiskAssessment()	Completes fall risk questionnaire

Caregiver (extends User)

Attribute	Type	Description
phoneNumber	String	Contact phone number
email	String	Email address

Method	Description
+manageMedicationSchedule()	Add/edit/delete medication schedules
+viewHealthDashboard()	View health analytics and logs
+receiveEmergencyAlert()	Receive SOS notifications

CaregiverDashboard

Attribute	Type	Description
calendarView	Calendar	Medication adherence calendar
medicationAdherenceData	Map<Date, Enum>	Daily adherence status
moodTrendGraph	Graph	Mood trend visualization

Method	Description
+displayCalendar()	Shows medication calendar
+updateMedicationSchedule(medID, newTime)	Updates schedule remotely
+visualizeMoodTrend()	Displays mood trend graph
+showAdherenceSummary()	Shows adherence statistics

ChatSession

Attribute	Type	Description
sessionID	UUID	Unique session identifier
conversationHistory	List<String>	History of conversation
detectedSentiment	String	Detected mood from conversation

Method	Description
+initiateGreeting()	Starts conversation with greeting
+processAudioInput()	Processes voice input via STT
+generateResponse()	Generates AI response

Medication

Attribute	Type	Description
medID	UUID	Unique medication identifier
name	String	Name of medication
dosage	String	Dosage information
scheduledTime	Time	Scheduled reminder time

Attribute	Type	Description
pillPhoto	Image	Photo of the pill
currentStatus	Enum	PENDING/TAKEN/SKIPPED/MISSED

Method	Description
+triggerReminderAlert()	Triggers medication reminder
+playAudioAnnouncement()	Plays voice reminder

MedicationLog

Attribute	Type	Description
logID	UUID	Unique log identifier
timestamp	DateTime	Time of action
statusAction	Enum	TAKEN/SKIPPED/MISSED

Method	Description
+logEvent()	Records medication event

EmergencyAlert

Attribute	Type	Description
alertID	UUID	Unique alert identifier
timestamp	DateTime	Time alert was triggered
gpsLocation	String	GPS coordinates

Method	Description
+initiateCountdown()	Starts 5-second countdown
+sendToCaregiver()	Sends alert to caregiver

FallRiskAssessment

Attribute	Type	Description
questionnaireData	List<Boolean>	Responses to questions

Attribute	Type	Description
calculatedRiskScore	Integer	Risk score (0-10)

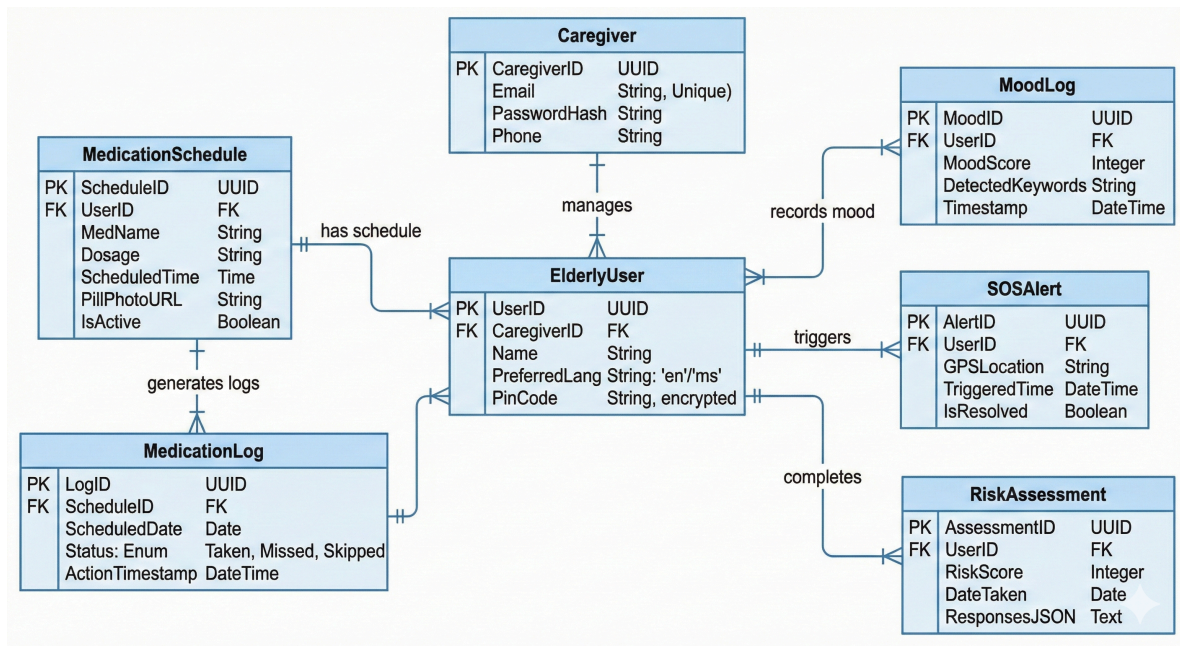
Method	Description
+presentQuestionnaire()	Displays questions
+calculateScore()	Calculates risk score
+displayRecommendation()	Shows recommendations

Class Relationships:

1. **User** is the parent class of **Elderly** and **Caregiver** (Inheritance)
 2. **Elderly** has 1 **Caregiver** who monitors them (1:1.* Association)
 3. **Elderly** participates in 0..* **ChatSession** (Association)
 4. **Elderly** has schedule of 0..* **Medication** (Composition)
 5. **Medication** generates 0..* **MedicationLog** (Association)
 6. **Elderly** triggers 0..* **EmergencyAlert** (Association)
 7. **Elderly** completes 0..* **FallRiskAssessment** (Association)
 8. **Caregiver** accesses 1 **CaregiverDashboard** (Association)
-

4.2 Information Viewpoint

Figure 2 – ER Diagram



4.2.1 Design Concern

This viewpoint aims to show the way that the system stores, manages, and manipulates the persistent information that the system will maintain.

4.2.2 Design Elements - Database Schema

Table: Caregiver

Column	Type	Constraints	Description
CaregiverID	UUID	PK	Primary key
Email	String	Unique	Login email
PasswordHash	String		Encrypted password
Phone	String		Contact number

Table: ElderlyUser

Column	Type	Constraints	Description
UserID	UUID	PK	Primary key
CaregiverID	UUID	FK	Foreign key to Caregiver
Name	String		User's name
PreferredLang	String		'en' or 'ms'

Column	Type	Constraints	Description
PinCode	String	Encrypted	Settings protection PIN

Table: MedicationSchedule

Column	Type	Constraints	Description
ScheduleID	UUID	PK	Primary key
UserID	UUID	FK	Foreign key to ElderlyUser
MedName	String		Medication name
Dosage	String		Dosage amount
ScheduledTime	Time		Reminder time
PillPhotoURL	String		URL to pill image
IsActive	Boolean		Active status

Table: MedicationLog

Column	Type	Constraints	Description
LogID	UUID	PK	Primary key
ScheduleID	UUID	FK	Foreign key to MedicationSchedule
ScheduledDate	Date		Date of scheduled dose
Status	Enum		TAKEN/MISSED/SKIPPED
ActionTimestamp	DateTime		Time of user action

Table: MoodLog

Column	Type	Constraints	Description
MoodID	UUID	PK	Primary key
UserID	UUID	FK	Foreign key to ElderlyUser
MoodScore	Integer		Score 0-10
DetectedKeywords	String		Keywords from chat
Timestamp	DateTime		Time recorded

Table: SOSAlert

Column	Type	Constraints	Description
AlertID	UUID	PK	Primary key
UserID	UUID	FK	Foreign key to ElderlyUser
GPSLocation	String		Coordinates
TriggeredTime	DateTime		Time triggered
IsResolved	Boolean		Resolution status

Table: RiskAssessment

Column	Type	Constraints	Description
AssessmentID	UUID	PK	Primary key
UserID	UUID	FK	Foreign key to ElderlyUser
RiskScore	Integer		Calculated score
DateTaken	Date		Assessment date
ResponsesJSON	Text		JSON of responses

Database Relationships:

- Caregiver **manages** ElderlyUser (1:N)
 - ElderlyUser **has schedule** MedicationSchedule (1:N)
 - MedicationSchedule **generates logs** MedicationLog (1:N)
 - ElderlyUser **records mood** MoodLog (1:N)
 - ElderlyUser **triggers** SOSAlert (1:N)
 - ElderlyUser **completes** RiskAssessment (1:N)
-

4.3 Interface Viewpoint

This section describes the interface and components of the SamaRasa application.

4.3.1 Design Concern

Concerns of the stakeholders in this document are **accessibility**, **simplicity**, and **reliability**. The UI must comply with WCAG 2.1 AAA standards.

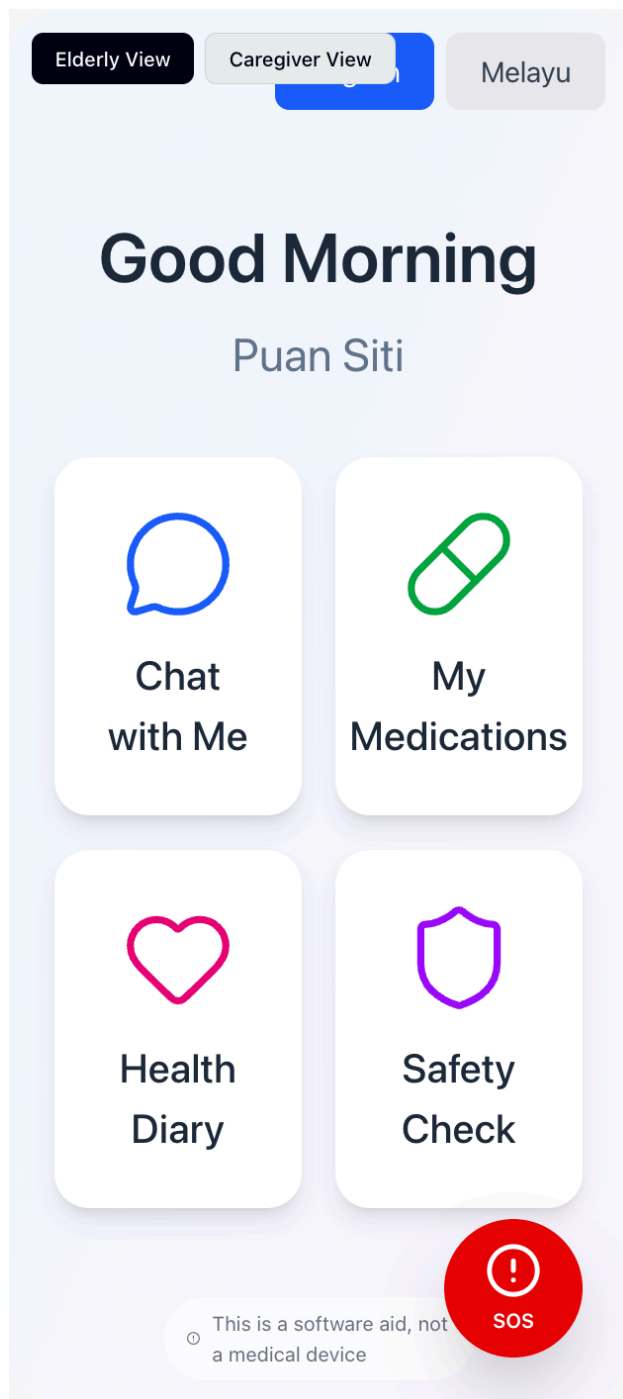
4.3.2 Design Elements - User Interface

UI Design Standards:

Element	Specification
Font Family	San Francisco / Roboto (Sans-serif)
Min Body Font Size	24px
Header Font Size	36px
Contrast Ratio	> 7:1 (WCAG AAA)
Touch Target Size	Minimum 48x48px
Color Coding	Green = Confirm, Red = Emergency/Stop, Blue = Information

4.3.2.1 Elderly View - Home Screen (English)

Figure 5 – Elderly View Home Screen (English)



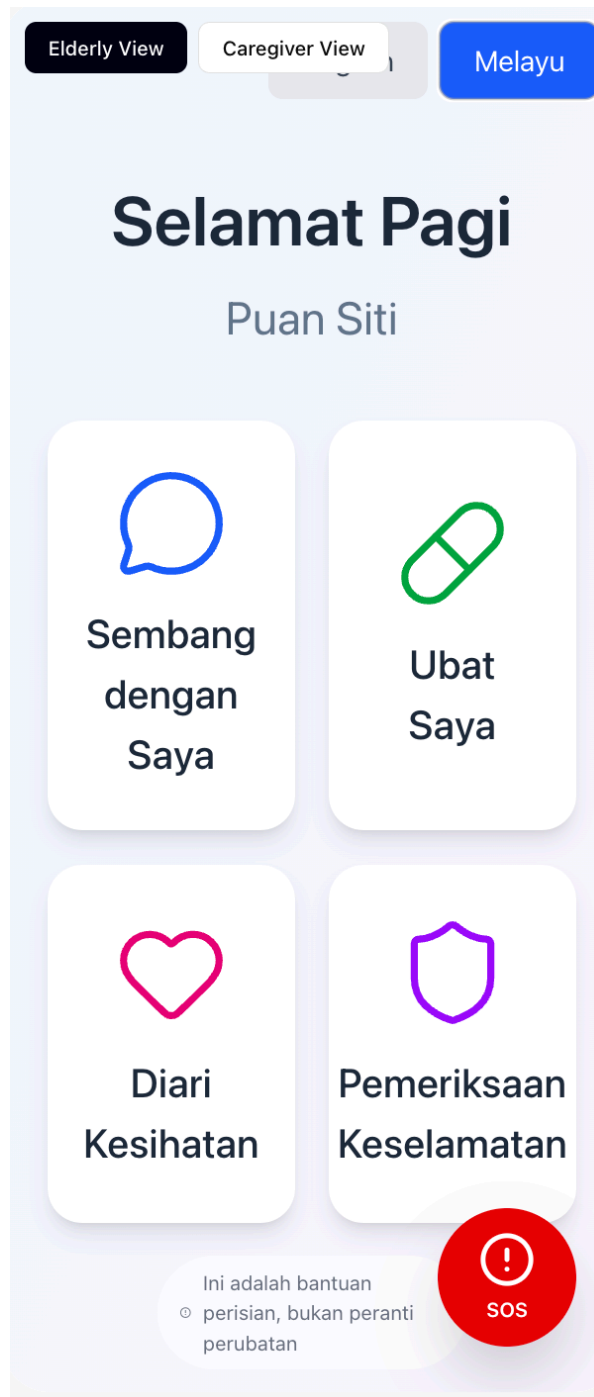
The home screen displays:

- **Greeting Message:** "Good Morning" (time-based)
- **User Name:** "Puan Siti"
- **Four Main Feature Cards** (2x2 grid):
 1. **Chat with Me** - Opens AI companion chat
 2. **My Medications** - Shows medication schedule

3. **Health Diary** - Opens mood logging
 4. **Safety Check** - Opens fall risk assessment
- **SOS Button:** Fixed red button at bottom-right
 - **Disclaimer:** "This is a software aid, not a medical device"
-

4.3.2.2 Elderly View - Home Screen (Malay)

Figure 6 – Elderly View Home Screen (Malay)



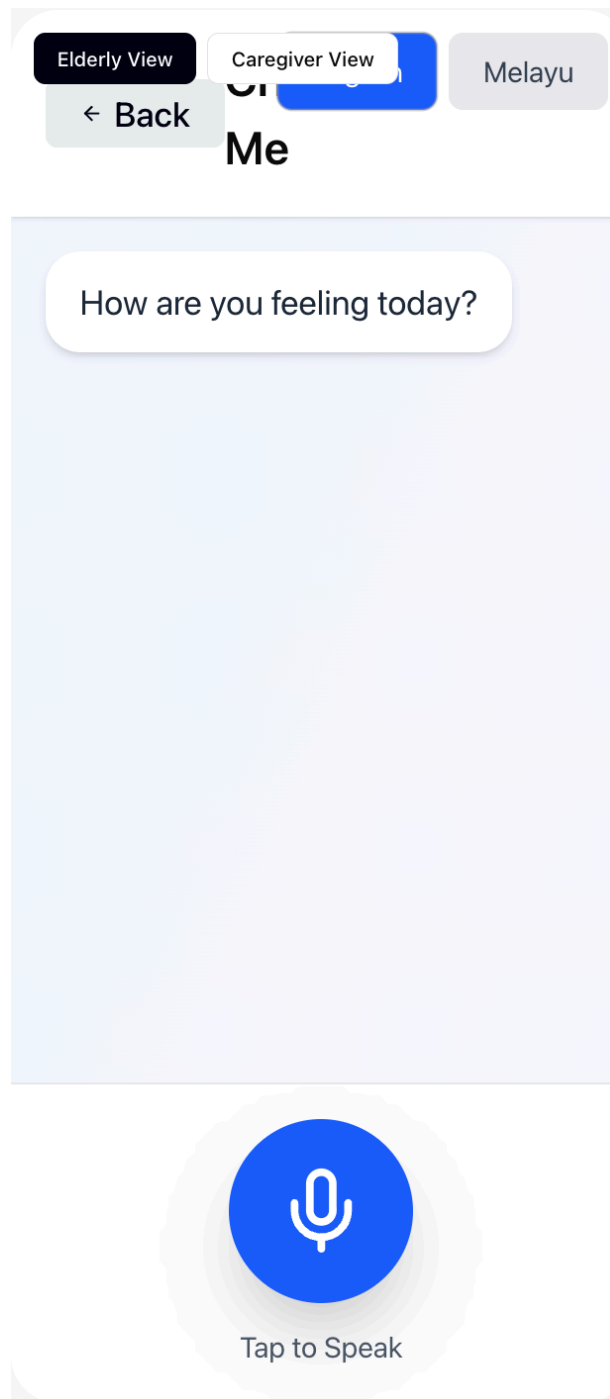
Same layout with Malay translations:

- "Selamat Pagi" (Good Morning)
- "Sembang dengan Saya" (Chat with Me)
- "Ubat Saya" (My Medications)
- "Diari Kesihatan" (Health Diary)
- "Pemeriksaan Keselamatan" (Safety Check)

Language toggle buttons at top: [English] [Melayu]

4.3.2.3 Chat with Me Screen

Figure 7 – Chat with Me Screen



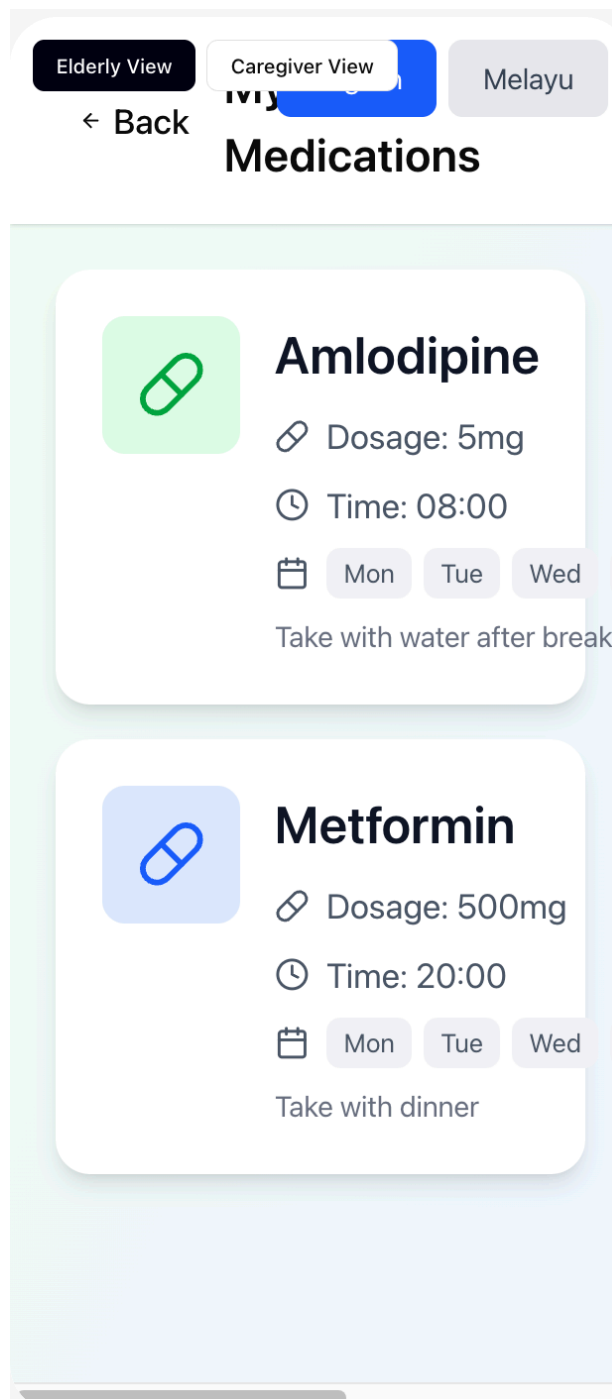
Interface elements:

- **Back Button:** "← Back" at top-left

- **Title:** "Chat with Me"
 - **AI Message Bubble:** "How are you feeling today?"
 - **Voice Input Button:** Large blue circular button with microphone icon
 - **Label:** "Tap to Speak"
 - **Visual Cue:** Pulsing waveform animation when listening
-

4.3.2.4 My Medications Screen

Figure 8 – My Medications Screen



Displays list of medications as cards:

Medication Card 1:

- Pill Icon (blue)
- **Name:** "Amlodipine"
- **Dosage:** 5mg
- **Time:** 08:00

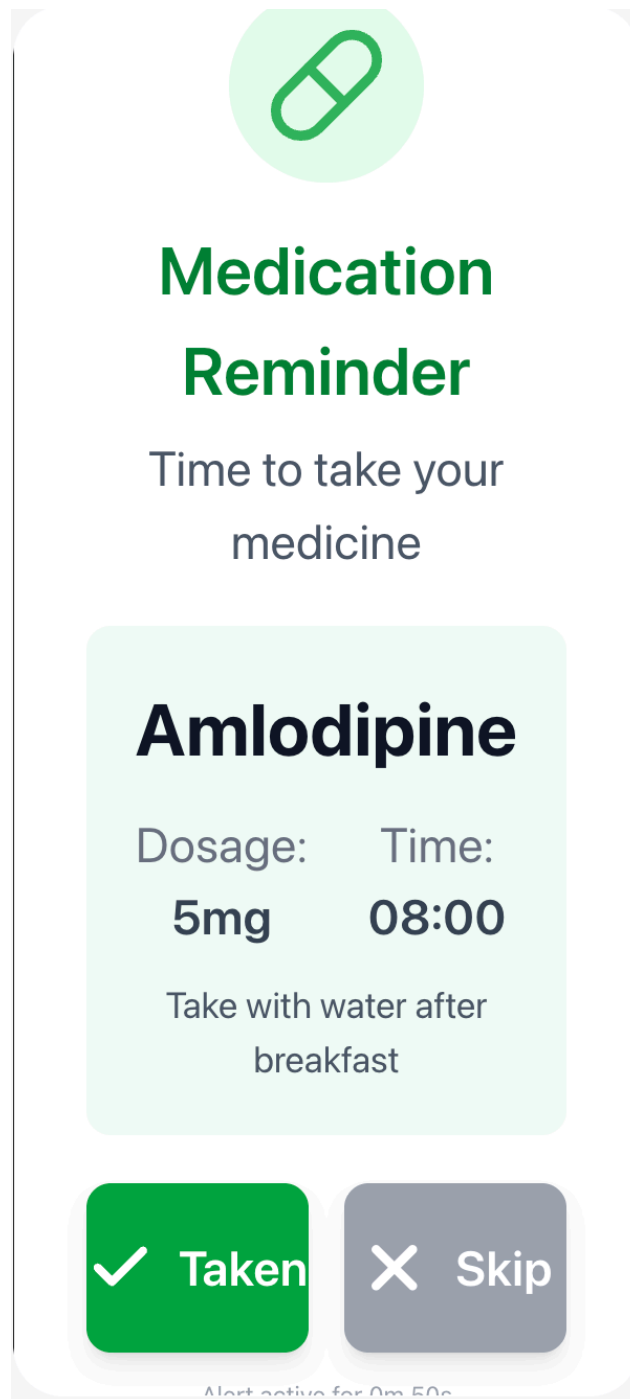
- **Days:** Mon, Tue, Wed (highlighted tags)
- **Instructions:** "Take with water after breakfast"

Medication Card 2:

- **Name:** "Metformin"
 - **Dosage:** 500mg
 - **Time:** 20:00
 - **Days:** Mon, Tue, Wed
 - **Instructions:** "Take with dinner"
-

4.3.2.5 Medication Reminder Alert Screen

Figure 9 – Medication Reminder Screen



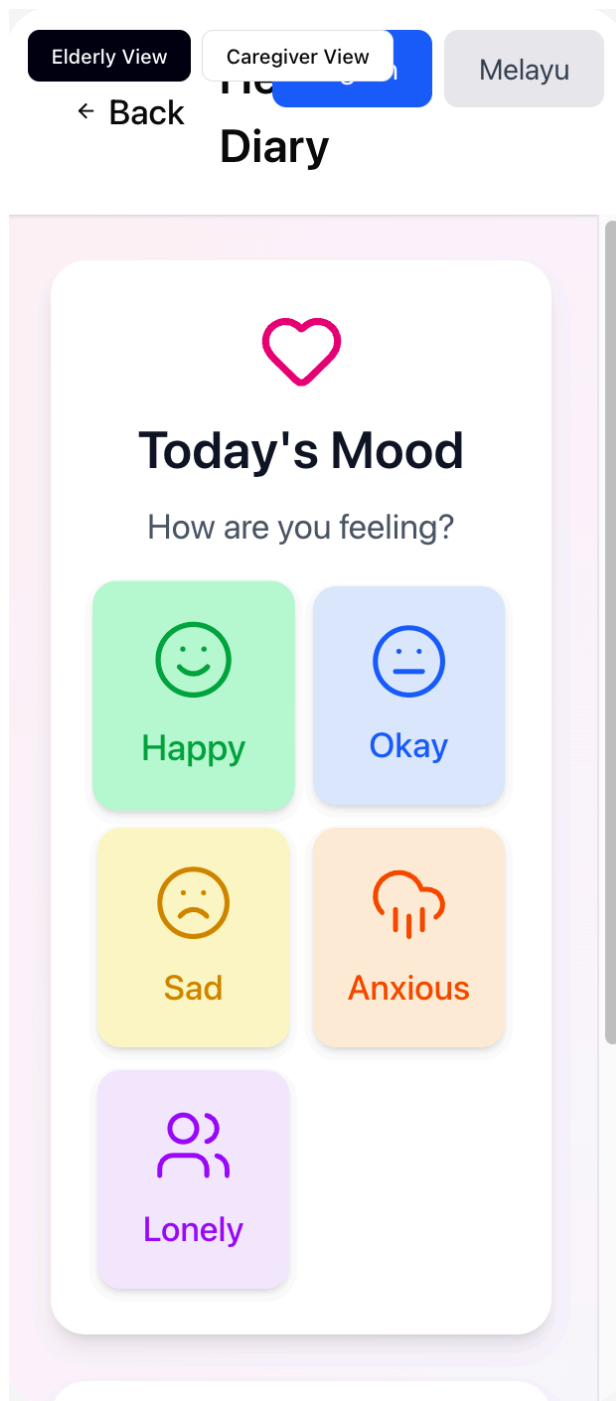
Full-screen modal alert:

- **Icon:** Large pill icon (green background)
- **Title:** "Medication Reminder"
- **Subtitle:** "Time to take your medicine"
- **Card:**
 - Medication Name: "Amlodipine"

- Dosage: 5mg | Time: 08:00
 - Instruction: "Take with water after breakfast"
 - **Action Buttons:**
 - [✓ Taken] - Large green button
 - [× Skip] - Gray button
 - **Timer:** "Alert active for: 0m 50s"
-

4.3.2.6 Health Diary Screen

Figure 10 – Health Diary Screen

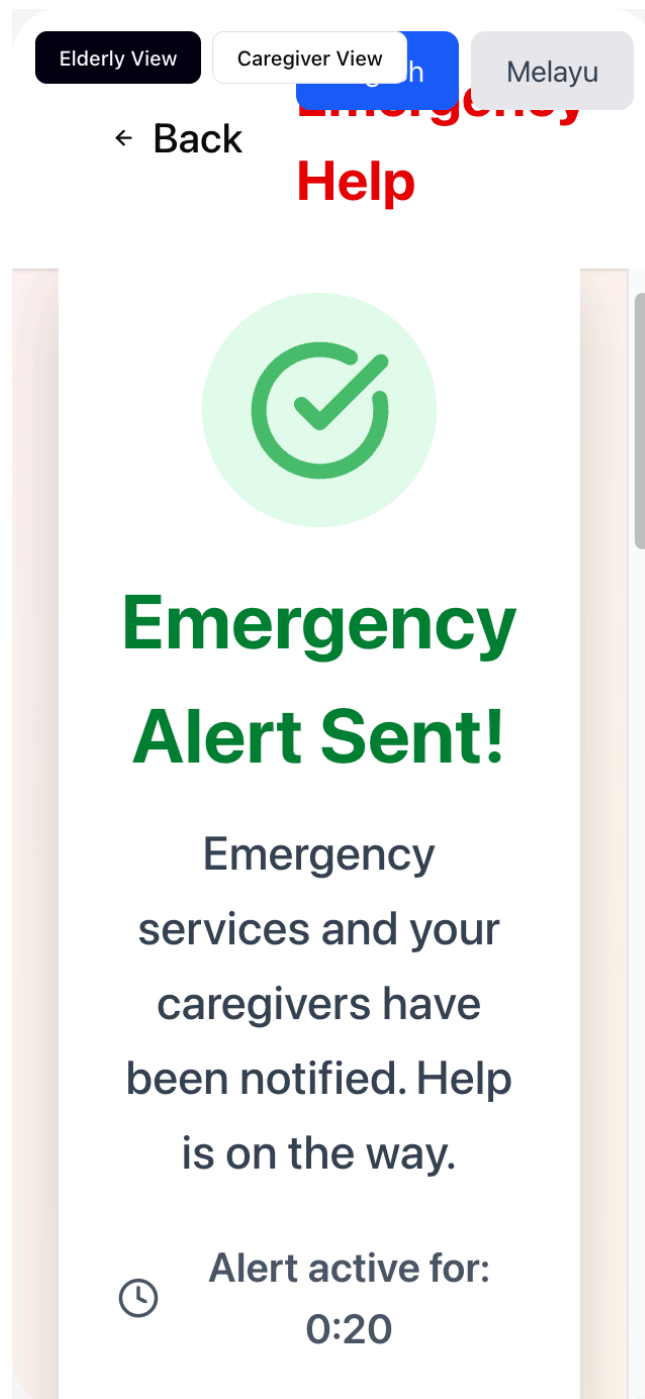


- **Title:** "Health Diary"
- **Card:** "Today's Mood - How are you feeling?"
- **Mood Options** (2x2 + 1 grid):
 - 😊 Happy (green)
 - 😐 Okay (blue)
 - 😞 Sad (yellow)

- 🤔 Anxious (orange)
- 👤👤 Lonely (purple)

4.3.2.7 Emergency Help Screen

Figure 11 – Emergency Help Screen



After SOS activation:

- **Icon:** Large green checkmark
 - **Title:** "Emergency Alert Sent!" (red text)
 - **Message:** "Emergency services and your caregivers have been notified. Help is on the way."
 - **Timer:** "Alert active for: 0:20"
-

4.3.2.8 Caregiver Login Screen

Figure 12 – Caregiver Login Screen

Elderly View

Caregiver View



SamaRasa

Caregiver Dashboard

Email

caregiver@example.com

Password

.....

→] Login

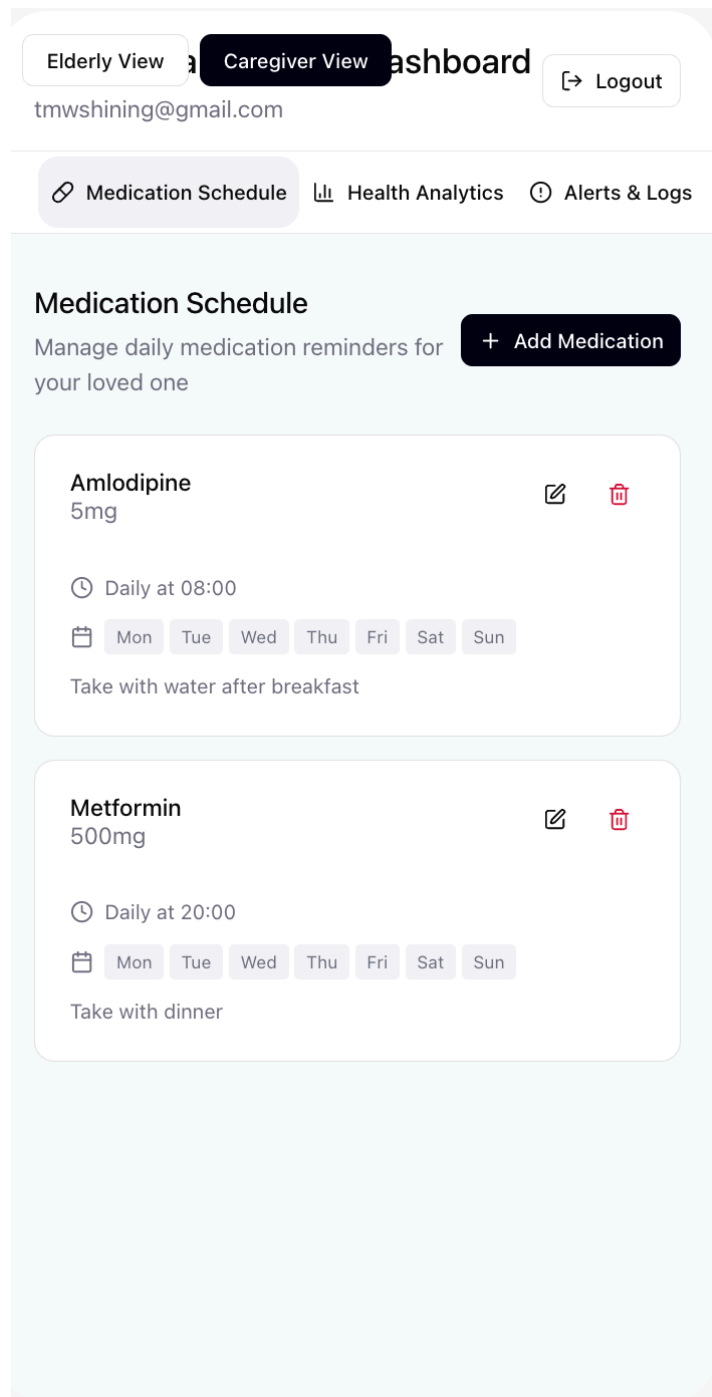
Demo login: Use any email and password

- **Logo:** Heart icon with "SamaRasa"
- **Subtitle:** "Caregiver Dashboard"
- **Form Fields:**
 - Email: caregiver@example.com
 - Password:
- **Login Button:** [→] Login

- **Helper Text:** "Demo login: Use any email and password"

4.3.2.9 Caregiver Dashboard - Medication Schedule

Figure 13 – Caregiver Dashboard - Medication Schedule



Header:

- "SamaRasa Dashboard" | [→ Logout]

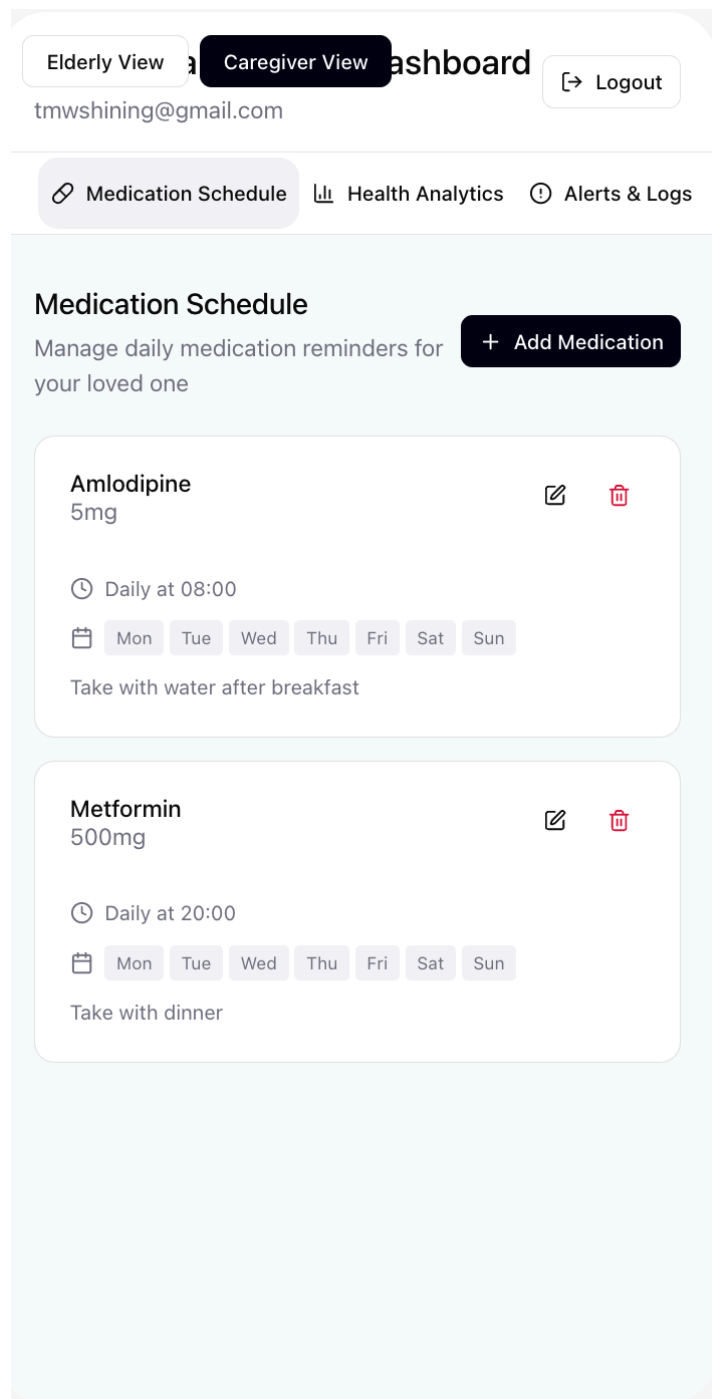
- User email display
- Tab Navigation: [Medication Schedule] [Health Analytics] [Alerts & Logs]

Content:

- **Title:** "Medication Schedule"
 - **Subtitle:** "Manage daily medication reminders for your loved one"
 - **[+ Add Medication]** button
 - **Medication Cards** with Edit/Delete icons:
 - Amlodipine 5mg - Daily at 08:00 - Mon-Sun - "Take with water after breakfast"
 - Metformin 500mg - Daily at 20:00 - Mon-Sun - "Take with dinner"
-

4.3.2.10 Caregiver Dashboard - Health Analytics

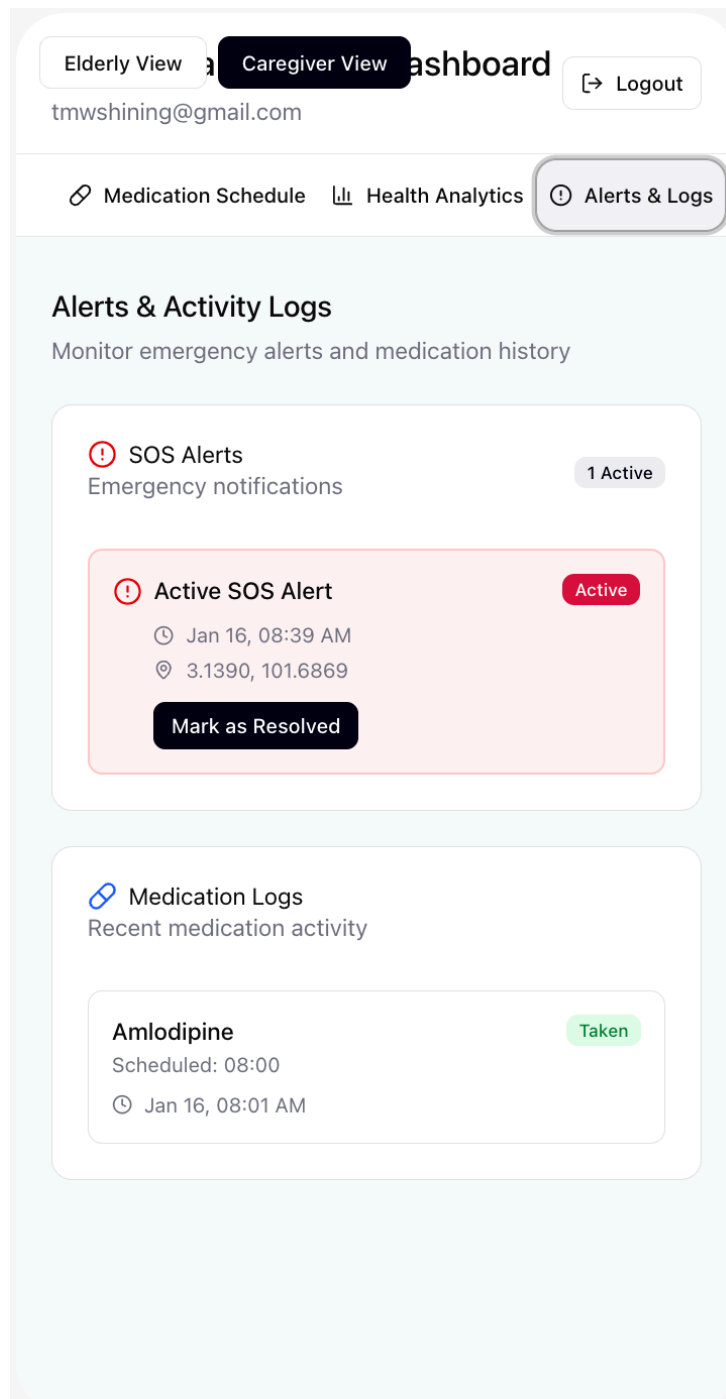
Figure 14 – Caregiver Dashboard - Health Analytics



- **Chart 1:** "Daily tracking of medication intake"
 - Bar chart showing Missed (red), Skipped (yellow), Taken (green)
 - X-axis: Dates (12/20 - 1/16)
- **Chart 2:** "Mood Trend (Last 14 Days)"
 - Line graph showing emotional well-being
 - Y-axis: Mood Score (0-10)

4.3.2.11 Caregiver Dashboard - Alerts & Logs

Figure 15 – Caregiver Dashboard - Alerts & Logs

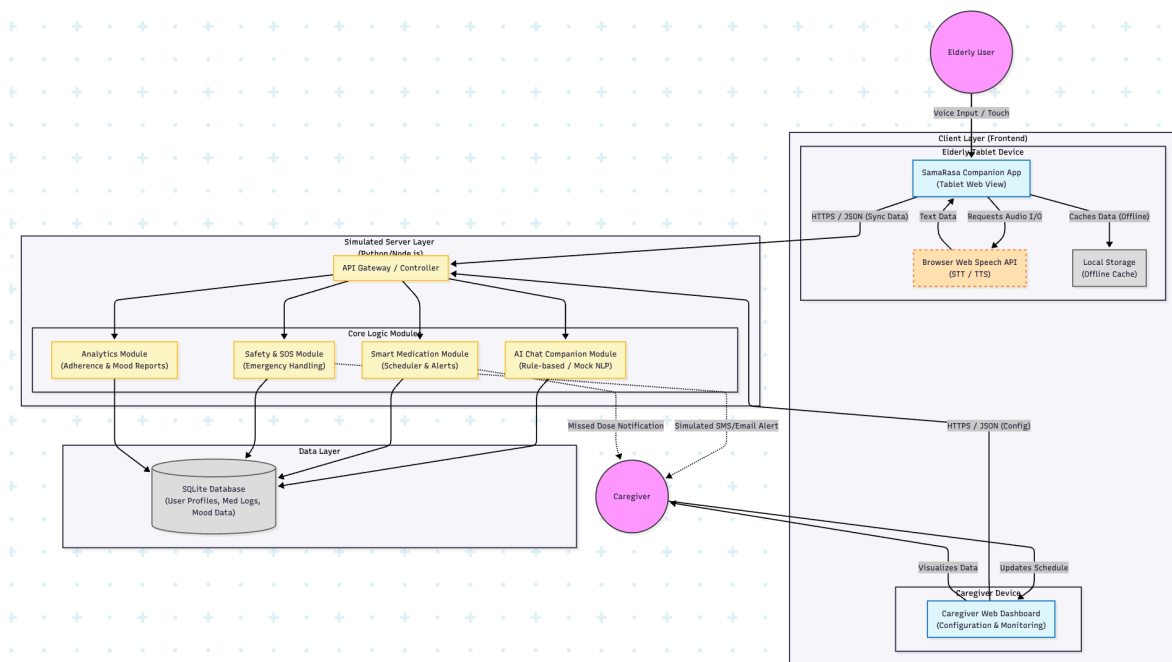


- **Section 1:** "SOS Alerts - Emergency notifications"
 - Alert Card: "Active SOS Alert" [Active badge]
 - Time: Jan 16, 08:39 AM

- Location: 3.1390, 101.6869
- [Mark as Resolved] button
- **Section 2: "Medication Logs - Recent medication activity"**
 - Log Entry: "Amlodipine - Scheduled: 08:00 - Jan 16, 08:01 AM" [Taken badge]

4.3.2.12 System Architecture Diagram

Figure 3 – System Architecture Diagram



The deployment architecture shows:

Client Layer (Frontend):

- Elderly's tablet device running SamaRasa Companion Web App
- Uses Browser Web Speech API (STT/TTS)
- Local Storage for offline cache

Simulated Server Layer:

- API Gateway / Controller (Node.js)
- Core Logic Modules:
 - Analytics Module (Adherence & Mood Reports)
 - Safety & SOS Module (Emergency Handling)

- Smart Medication Module (Scheduler & Alerts)
- AI Chat Companion Module (Rule-based / Mock NLP)
- Data Layer: SQLite Database

Caregiver Device:

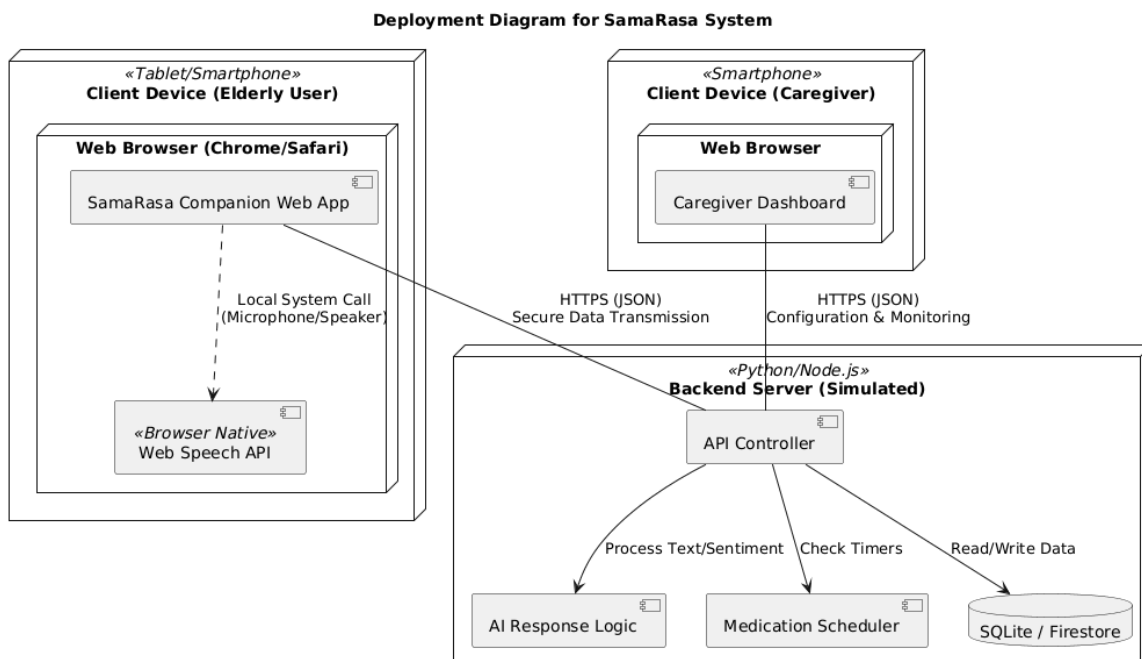
- Web Browser accessing Caregiver Dashboard
- HTTPS/JSON communication with backend

Communication:

- HTTPS / JSON (Secure Data) between all components
- Missed Dose Notifications and SOS Alerts via SMS/Email simulation

4.3.2.13 Deployment Diagram

Figure 4 – Deployment Diagram

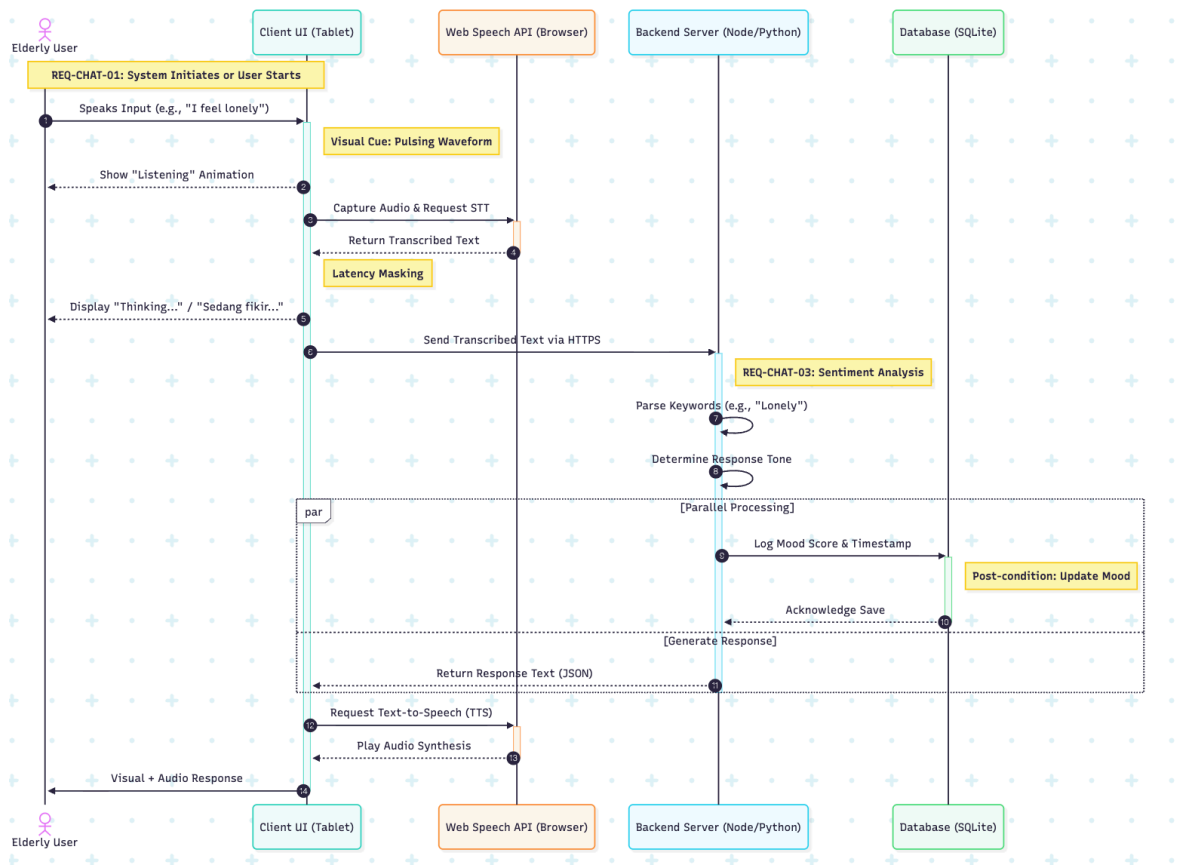


4.4 Interaction Viewpoint

This section presents sequence diagrams showing how components interact during key use cases.

4.4.1 AI Chat Companion Component

Figure 16 – Sequence Diagram of AI Chat Component



Scenario: Daily Greeting & Mood Capture (REQ-CHAT-01)

Actors/Objects: Elderly User → Client UI (Tablet) → Web Speech API (Browser) → Backend Server (Node/Python) → Database (SQLite/Firebase)

Flow:

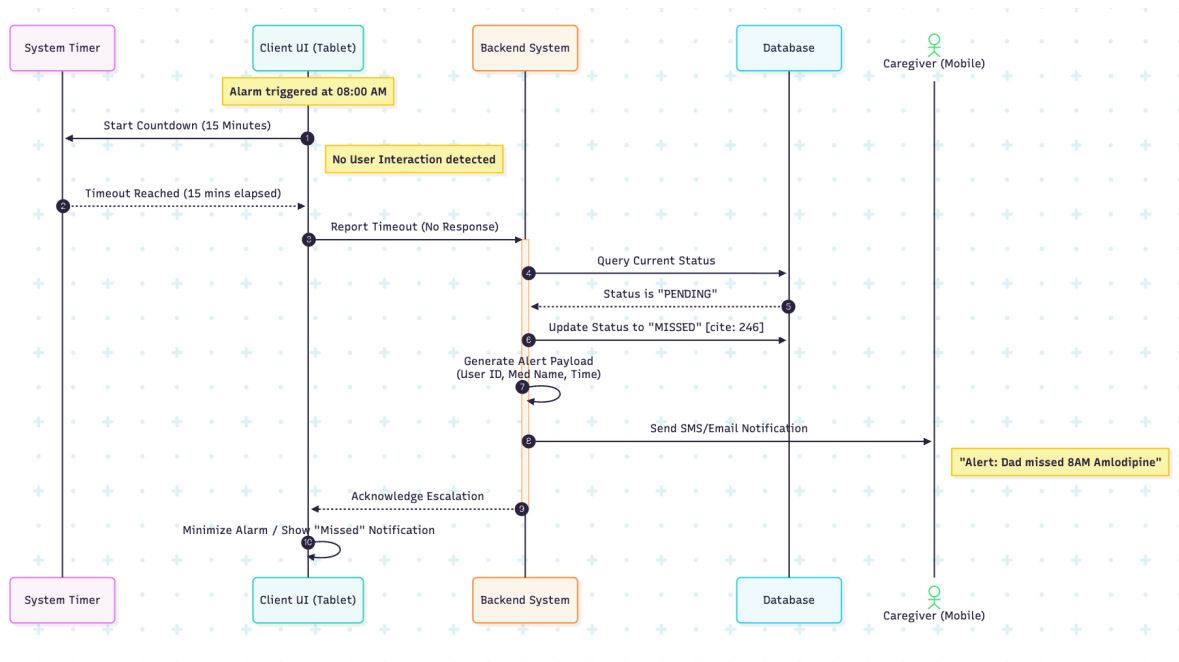
1. System triggers greeting based on time ("Good Morning")
2. Client UI requests TTS (Text-to-Speech)
3. Audio plays: "Good morning! How are you feeling today?"
4. User speaks (e.g., "I feel a bit sad and dizzy")
5. Client shows "Listening" animation (pulsing waveform)
6. Audio stream sent to Web Speech API
7. Transcribed text returned: "I feel a bit sad and dizzy"
8. Client displays "Thinking..." / "Sedang fikir..."
9. POST /api/conversation (text, userID) to backend

10. Backend parses keywords (Sentiment Analysis - REQ-CHAT-03):
 - Detected: "Sad" (Mood), "Dizzy" (Health Symptom)
11. INSERT INTO DailyLogs (MoodScore, Keywords, Time)
12. Database acknowledges save
13. Backend generates empathetic response
14. Return JSON: {response: "I understand. Please rest...", tone: "soft"}
15. Client requests TTS for response
16. Audio output plays (empathetic voice)
17. Visual + Audio response shown to user

Post-condition: Updates Mood Trend for Caregiver Dashboard

4.4.2 Medication Reminder Component

Figure 17 – Sequence Diagram of Medication Reminder Component



Actors/Objects: System Timer → Client UI (Tablet) → Backend System → Database → Caregiver (Mobile)

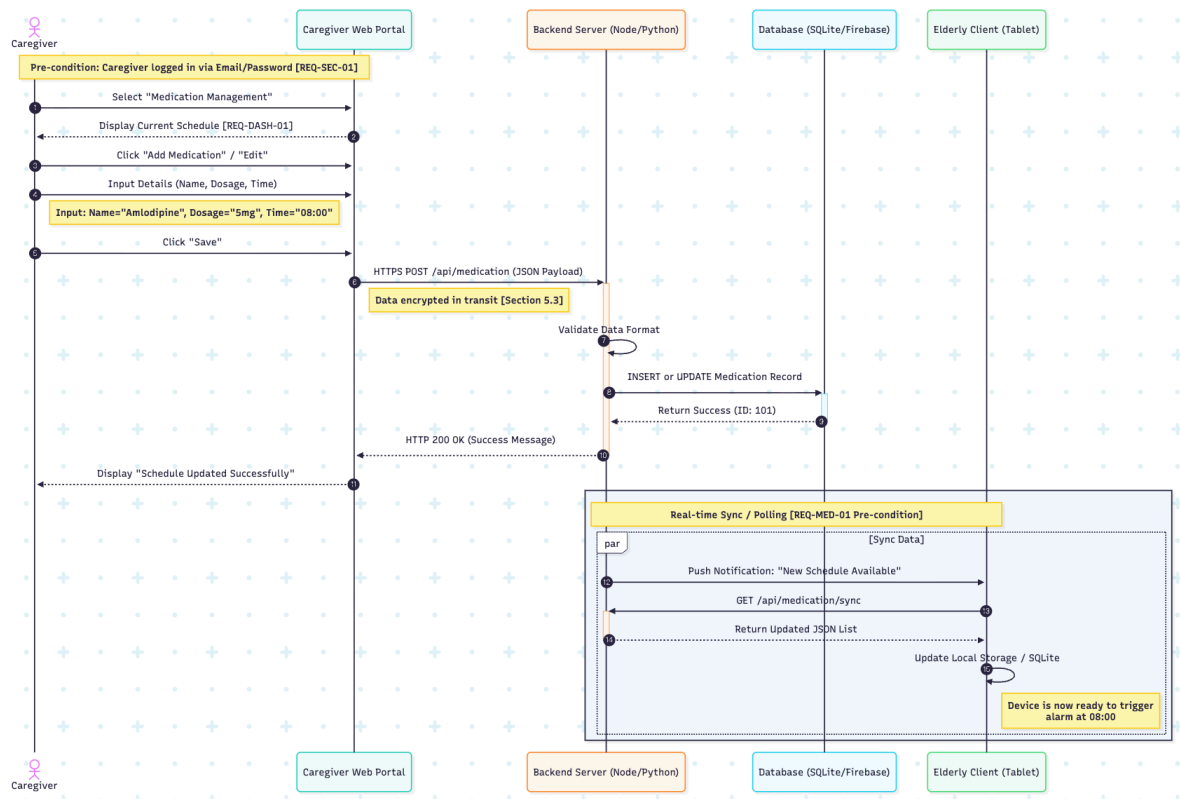
Flow:

1. Alarm triggered at 08:00 AM

2. Full-screen modal displayed (REQ-MED-01)
 3. System starts countdown (15 minutes)
 4. Audio announcement plays every 30 seconds (REQ-MED-03)
 5. **If user responds:**
 - User clicks [Taken] or [Skip]
 - Status updated in database (REQ-MED-05)
 - Alarm dismissed
 6. **If no user interaction (timeout reached):**
 - Backend reports timeout
 - Query current status: "PENDING"
 - Update status to "MISSED"
 - Generate alert payload (User ID, Med Name, Time)
 - Send SMS/Email notification to caregiver (REQ-MED-04)
 - Caregiver receives: "Alert: Dad missed 8AM Amlodipine"
 7. Client minimizes alarm / Shows "Missed" notification
-

4.4.3 Caregiver Add Medication Component

Figure 18 – Sequence Diagram of Caregiver Add Medication Component



Pre-condition: Caregiver logged in via Email/Password (REQ-SEC-01)

Actors/Objects: Caregiver → Caregiver Web Portal → Backend Server → Database → Elderly Client (Tablet)

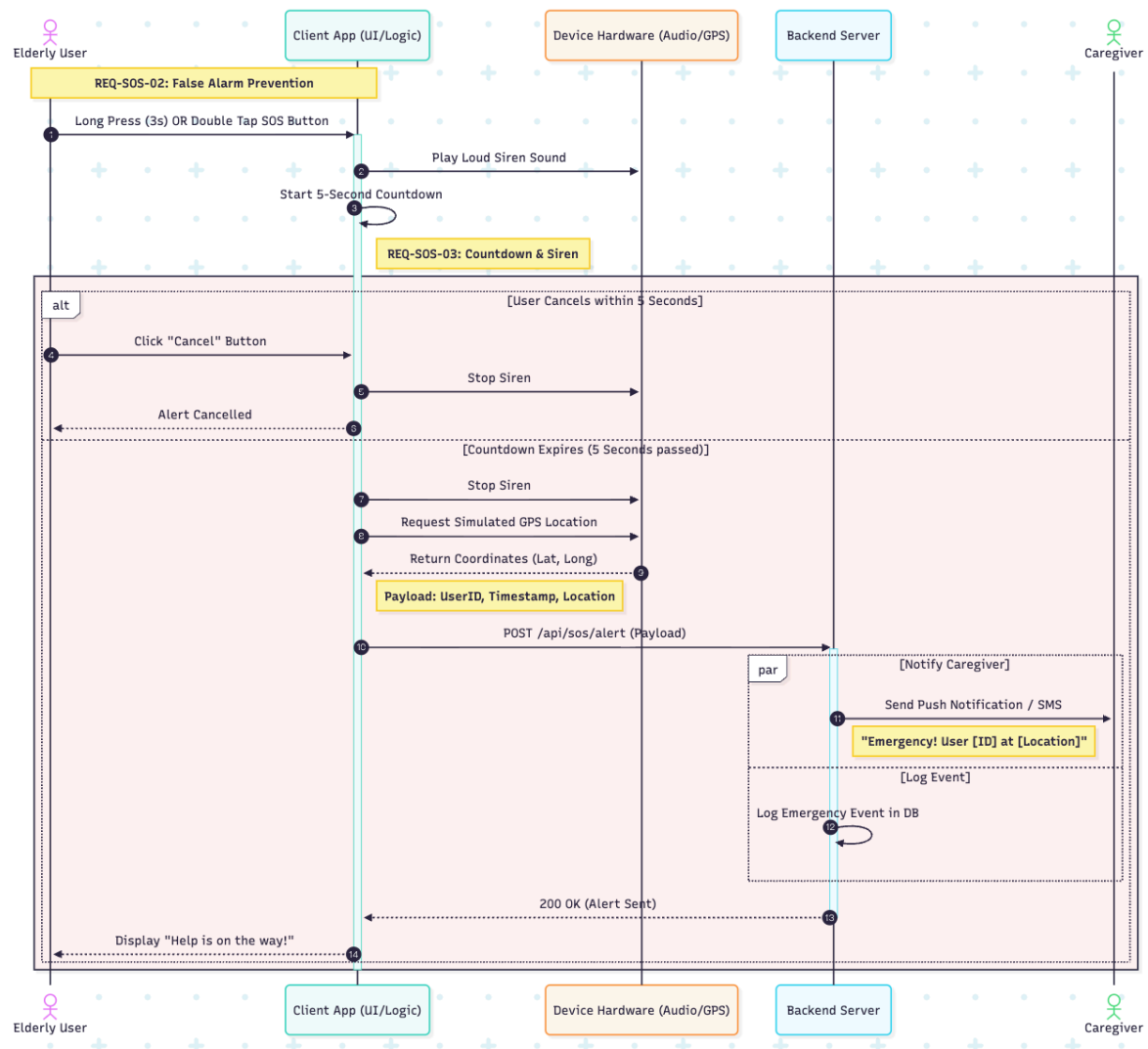
Flow:

1. Caregiver selects "Medication Management"
2. Portal displays current schedule (REQ-DASH-01)
3. Caregiver clicks [+ Add Medication] / [Edit]
4. Input details (Name, Dosage, Time)
5. Example input: Name="Amlodipine", Dosage="5mg", Time="08:00"
6. Caregiver clicks [Save]
7. HTTPS POST /api/medication (JSON Payload)
8. Data encrypted in transit
9. Backend validates data format
10. INSERT or UPDATE Medication Record
11. Return Success (ID: 101)

12. HTTP 200 OK (Success Message)
13. Portal displays "Schedule Updated Successfully"
14. **Real-time Sync** (REQ-MED-01 Pre-condition):
 - Backend pushes notification: "New Schedule Available"
 - Elderly device GET /api/medication/sync
 - Return updated JSON list
 - Update Local Storage / SQLite
15. Device is now ready to trigger alarm at 08:00

4.4.4 SOS Emergency Alert Component

Figure 19 – Sequence Diagram of SOS Component



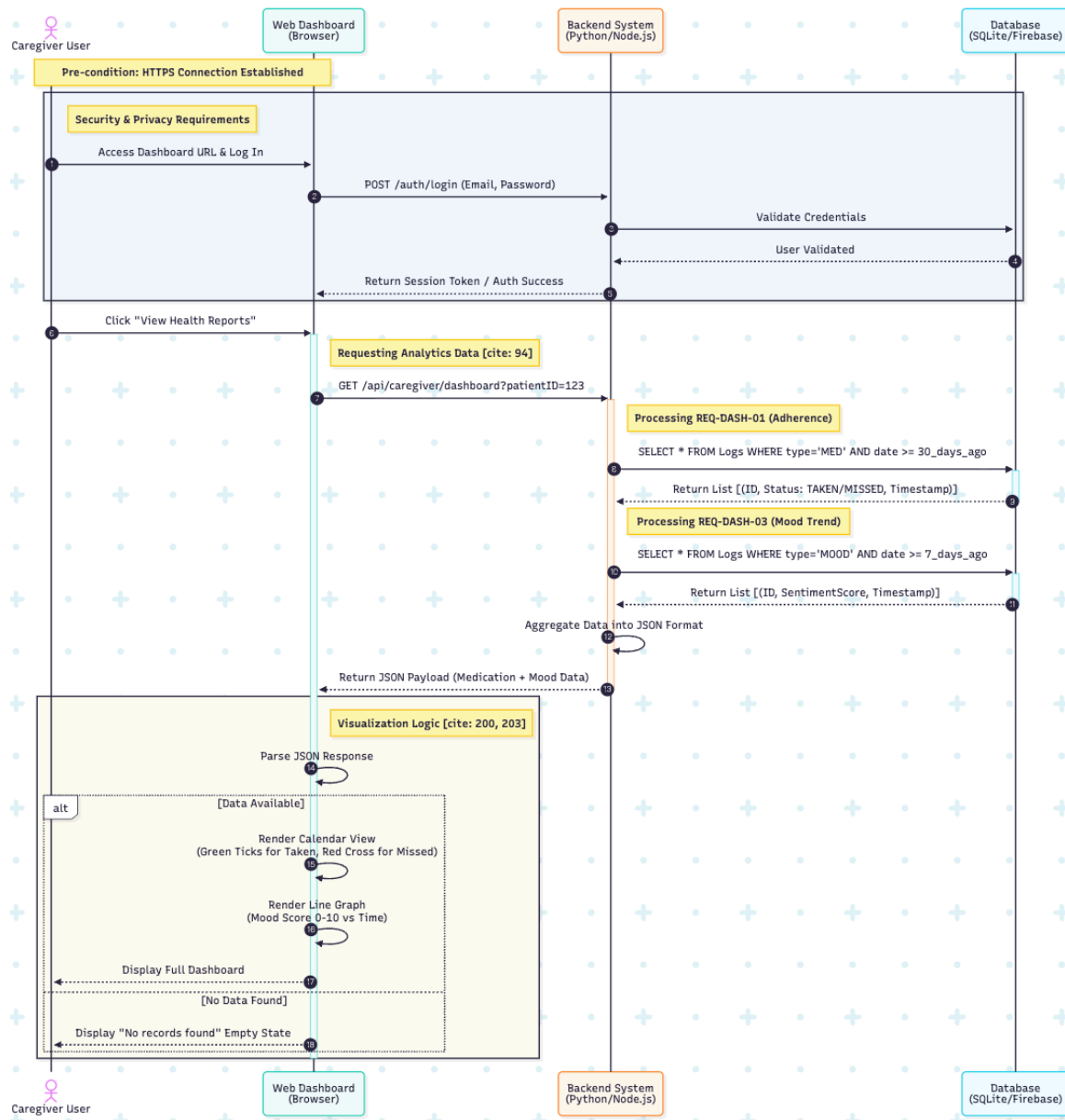
Actors/Objects: Elderly User → Client App (UI/Logic) → Device Hardware (Audio/GPS) → Backend Server → Caregiver

Flow (REQ-SOS-02: False Alarm Prevention):

1. User long presses (3s) OR double taps SOS button
2. Device plays loud siren sound
3. Client starts 5-second countdown (REQ-SOS-03)
4. **[User cancels within 5 seconds]:**
 - User clicks [Cancel] button
 - Stop siren
 - Alert cancelled
5. **[Countdown expires (5 seconds passed)]:**
 - Stop siren
 - Request simulated GPS location
 - Return coordinates (Lat, Long)
 - Build payload: UserID, Timestamp, Location (REQ-SOS-04)
 - POST /api/sos/alert (Payload)
 - **Parallel processing:**
 - Send push notification / SMS to caregiver
 - Caregiver receives: "Emergency! User [ID] at [Location]"
 - Log emergency event in database
 - Return 200 OK (Alert Sent)
 - Display "Help is on the way!"

4.4.5 Caregiver Dashboard - Health Analytics Component

Figure 20 – Sequence Diagram of Caregiver Dashboard Component



Pre-condition: HTTPS Connection Established, Security & Privacy Requirements met

Actors/Objects: Caregiver User → Web Dashboard (Browser) → Backend System (Python/Node.js) → Database (SQLite/Firebase)

Flow:

1. Caregiver accesses dashboard URL and logs in
2. POST /auth/login (Email, Password)
3. Backend validates credentials
4. User validated → Return session token
5. Caregiver clicks "View Health Reports"

6. **Requesting Analytics Data:**

- GET /api/caregiver/dashboard?patientID=123

7. **Processing REQ-DASH-01 (Adherence):**

- SELECT * FROM Logs WHERE type='MED' AND date >= 30_days_ago
- Return list [{ID, Status: TAKEN/MISSED, Timestamp}]

8. **Processing REQ-DASH-03 (Mood Trend):**

- SELECT * FROM Logs WHERE type='MOOD' AND date >= 7_days_ago
- Return list [{ID, SentimentScore, Timestamp}]

9. Backend aggregates data into JSON format

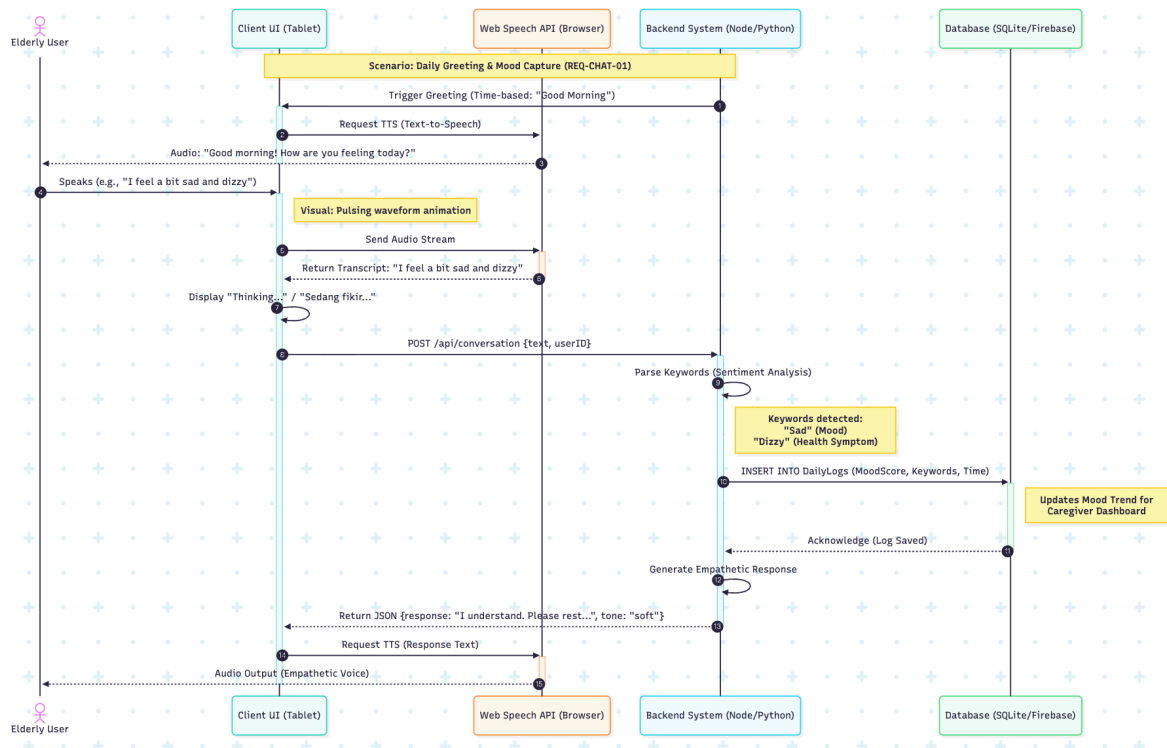
10. Return JSON payload (Medication + Mood Data)

11. **Visualization Logic:**

- Parse JSON response
- **[If data available]:**
 - Render calendar view (Green ticks for Taken, Red cross for Missed)
 - Render line graph (Mood Score 0-10 vs Time)
 - Display full dashboard
- **[If no data found]:**
 - Display "No records found" empty state

4.4.6 Elderly Mood Recording Component

Figure 21 – Sequence Diagram of Mood Recording Component



Scenario: Mood captured through AI conversation

Actors/Objects: Elderly User → Client UI (Tablet) → Web Speech API → Backend System → Database

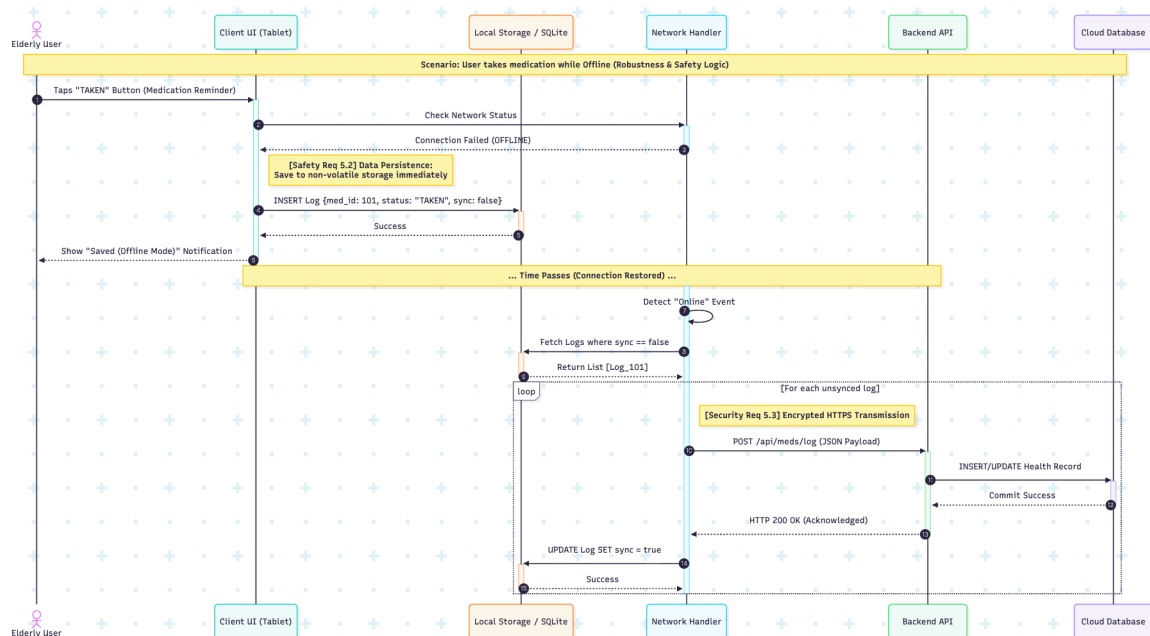
Flow:

1. Trigger greeting (time-based: "Good Morning")
2. Request TTS
3. Audio: "Good morning! How are you feeling today?"
4. User speaks: "I feel a bit sad and dizzy"
5. Visual: Pulsing waveform animation
6. Send audio stream to Web Speech API
7. Return transcript: "I feel a bit sad and dizzy"
8. Display "Thinking..." / "Sedang fikir..."
9. POST /api/conversation (text, userID)
10. Parse keywords (Sentiment Analysis):
 - Detected: "Sad" (Mood), "Dizzy" (Health Symptom)
11. INSERT INTO DailyLogs (MoodScore, Keywords, Time)

12. Updates Mood Trend for Caregiver Dashboard
13. Acknowledge (Log Saved)
14. Generate empathetic response
15. Return JSON: {response: "I understand. Please rest...", tone: "soft"}
16. Request TTS (Response Text)
17. Audio output (Empathetic voice)

4.4.7 Data Synchronization Component

Figure 22 – Sequence Diagram of Data Synchronization



Scenario: User takes medication while offline (Robustness & Safety Log)

Actors/Objects: Elderly User → Client UI (Tablet) → Local Storage / SQLite → Network Handler → Backend API → Cloud Database

Flow:

1. User taps "TAKEN" button (Medication Reminder)
2. Check network status
3. Connection failed (OFFLINE)
4. **Safety Req 5.2:** Data Persistence - Save to non-volatile storage immediately

5. INSERT Log (med_id: 101, status: "TAKEN", sync: false)
 6. Success acknowledgment
 7. Show "Saved (Offline Mode)" notification
 8. ... **Time passes (Connection Restored)** ...
 9. Network handler detects "Online" event
 10. Fetch logs where sync == false
 11. Return list [Log_101]
 12. **Loop** for each unsynced log:
 - **Security Req 5.3:** Encrypted HTTPS transmission
 - POST /api/meds/log (JSON Payload)
 - INSERT/UPDATE Health Record
 - Commit success
 - HTTP 200 OK (Acknowledged)
 - UPDATE Log SET sync = true
 - Success
-

4.5 Context Viewpoints

Use case diagrams showing system actors and their interactions.

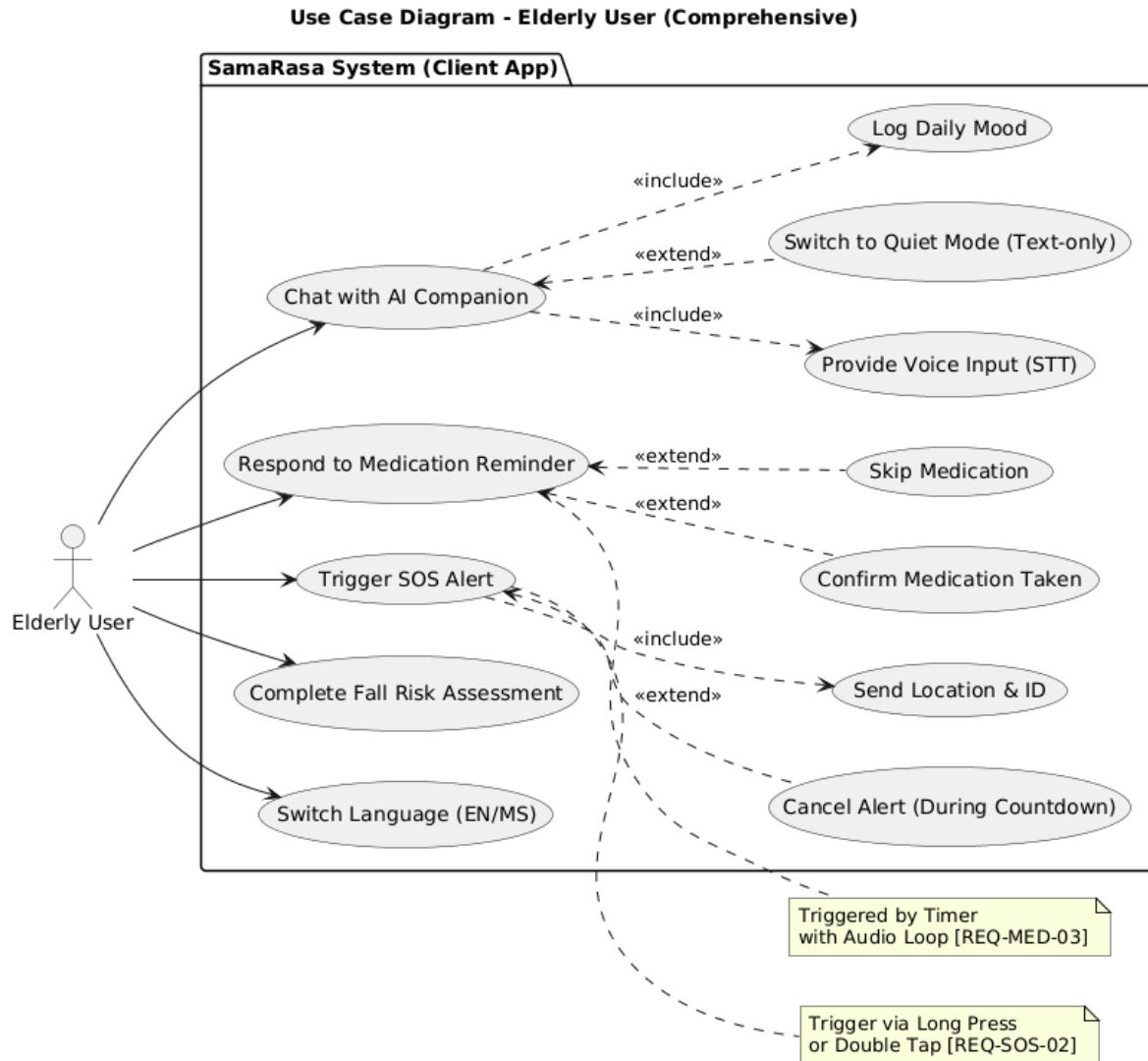
4.5.1 Design Concern

Concern of the context viewpoint is to determine properties of users and their interactions with the system. Clear definition of user capabilities ensures efficient implementation.

4.5.2 Design Elements - Use Case Diagrams

4.5.2.1 Use Case Diagram - Elderly User

Figure 23 – Use Case Diagram of Elderly User



Actor: Elderly User

System: SamaRasa System (Client App)

Use Cases:

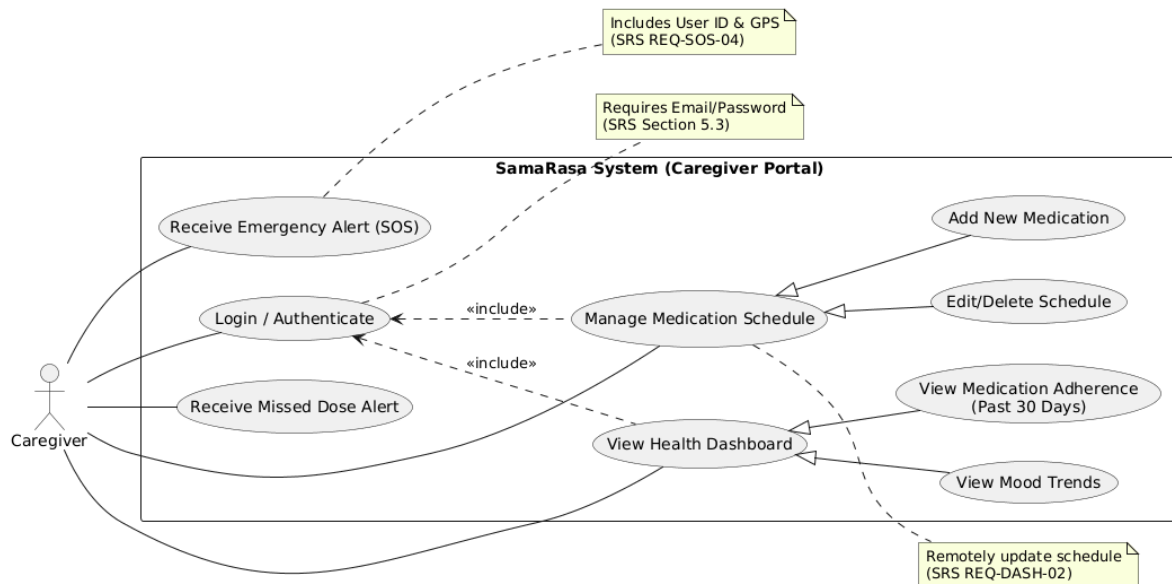
Use Case	Description	Relationships
Chat with AI Companion	Engage in conversation with AI	«include» Log Daily Mood, «include» Provide Voice Input (STT), «extend» Switch to Quiet Mode (Text-only)
Respond to Medication Reminder	React to medication alerts	«extend» Skip Medication, «extend» Confirm Medication Taken
Trigger SOS Alert	Activate emergency help	«include» Send Location & ID, «extend» Cancel Alert (During Countdown)
Complete Fall Risk Assessment	Answer safety questionnaire	-
Switch Language (EN/MS)	Toggle between languages	-
Log Daily Mood	Record emotional state	Triggered by Timer with Audio Loop [REQ-MED-03]

Notes:

- Medication Reminder: Triggered by Timer with Audio Loop [REQ-MED-03]
- SOS Alert: Trigger via Long Press or Double Tap [REQ-SOS-02]

4.5.2.2 Use Case Diagram - Caregiver

Figure 24 – Use Case Diagram of Caregiver



Actor: Caregiver

System: SamaRasa System (Caregiver Portal)

Pre-conditions:

- Requires Email/Password [SRS Section 5.3]
- Includes User ID & GPS [SRS REQ-SOS-04]

Use Cases:

Use Case	Description	Relationships
Login / Authenticate	Secure access to portal	«include» Manage Medication Schedule
Receive Emergency Alert (SOS)	Get notified of emergencies	-
Receive Missed Dose Alert	Get notified of missed medications	-
Manage Medication Schedule	Configure medications	«extend» Add New Medication, «extend» Edit/Delete Schedule
View Health Dashboard	Access health data	«extend» View Medication Adherence (Past 30 Days), «extend» View Mood Trends, «extend» Remotely update schedule [SRS REQ-DASH-02]

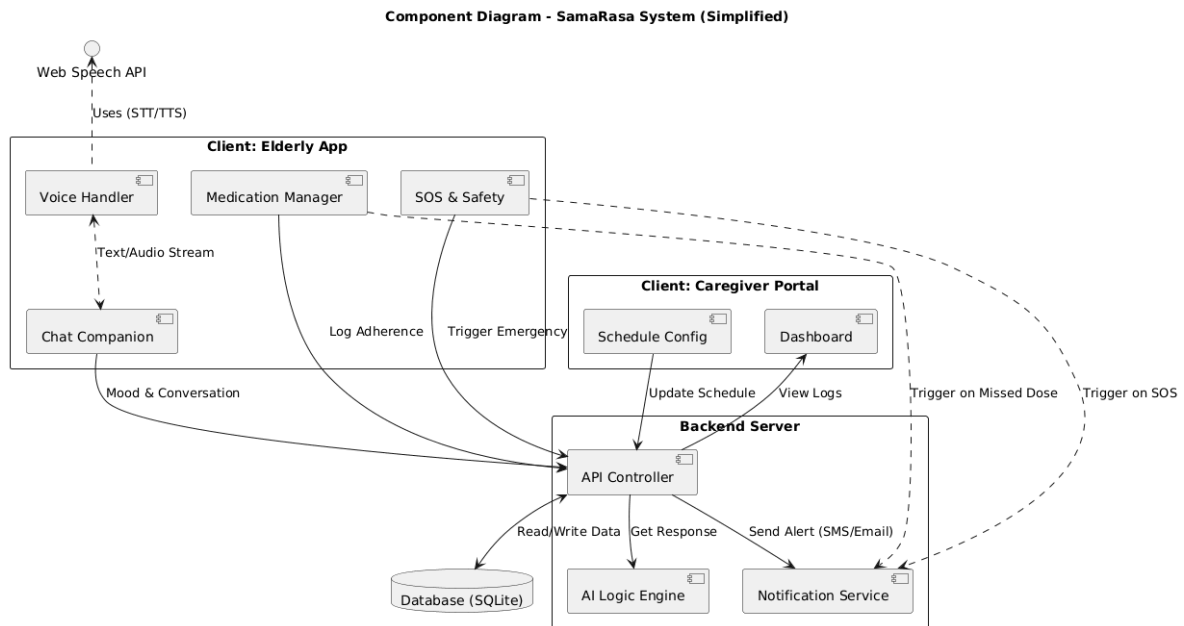
4.6 Composition Viewpoints

4.6.1 Design Concerns

This viewpoint states information related to planning, monitoring, and controlling the project. To give an early view of the logical software components, we provide a component diagram.

4.6.2 Design Elements - Component Diagram

Figure 25 – Component Diagram



Component Descriptions:

Component	Type	Description	Dependencies
Voice Handler	Module	Processes voice input/output	Web Speech API
Chat Companion	Module	Manages AI conversation and mood detection	Voice Handler, AI Logic Engine
Medication Manager	Module	Handles medication scheduling and reminders	Database, Notification Service
SOS & Safety	Module	Manages emergency alerts	API Controller, Notification Service
Schedule Config	Module	Allows caregivers to configure medications	API Controller, Database
Dashboard	Module	Displays health analytics and logs	API Controller, Database
API Controller	Component	Routes requests and handles business logic	All modules
Database (SQLite)	Data Store	Persistent storage for all data	-
AI Logic Engine	Service	Processes sentiment and generates responses	Database
Notification Service	Service	Sends SMS/Email alerts	-

5. Requirements Matrix

This matrix traces functional requirements from the SRS to design components.

UC = Use Case from SRS

Use Case ID	Description	Login	Chat	Medication	SOS	Dashboard	Risk Assessment
REQ-CHAT-01	Time-based greeting		X				
REQ-CHAT-02	Voice capture & STT		X				
REQ-CHAT-03	Sentiment analysis		X				
REQ-CHAT-04	Voice fallback (<60% confidence)		X				
REQ-CHAT-05	Quiet mode (text-only)		X				
REQ-MED-01	Full-screen reminder alert			X			
REQ-MED-02	Alert with pill info & buttons			X			
REQ-MED-03	Audio loop every 30s			X			
REQ-MED-04	Escalation after 15 min			X		X	
REQ-MED-05	Log medication status			X		X	
REQ-SOS-01	Fixed SOS button position				X		
REQ-SOS-02	Long press / double tap activation				X		
REQ-SOS-03	5-second countdown with siren				X		
REQ-SOS-04	Alert payload (ID, time, GPS)				X	X	
REQ-DASH-01	Adherence calendar view (30 days)	X				X	
REQ-DASH-02	Remote schedule management	X		X		X	
REQ-DASH-03	Mood trend visualization		X			X	
REQ-RISK-01	Questionnaire wizard						X
REQ-RISK-02	Risk score calculation						X
REQ-SEC-01	Caregiver authentication	X				X	

End of Document

Document Information:

- **Project:** SamaRasa - AI-Powered Companion System
- **Course:** SSW3001 Software Engineering
- **Semester:** 1 2025/2026
- **University:** Universiti Putra Malaysia
- **Submission:** Assignment 2 - Software Design Document (SDD)